

# Comparative Evaluation Report: Puzzle Game Implementations

## 1. Executive Summary

This report presents a comprehensive comparative evaluation of three distinct puzzle game implementations: the Sumlink Android Application, the Sumlink Web Application, and the Number Match Puzzle Game (Deterministic Version). The primary objective is to analyze performance, gameplay mechanics, difficulty scaling, and statistical reliability across the different platforms. By evaluating key metrics such as completion times, win rates, and player actions (moves and row additions), this report provides actionable insights into the strengths and weaknesses of each system, concluding with recommendations for future optimizations.

## 2. Testing Methodology

The data collection methodology varied significantly between the tested systems:

- Sumlink App (Android):** Data was collected through manual testing across 10 completed game sessions.
- Sumlink Web (Netlify):** Data was collected through manual testing across 6 completed game sessions.
- Number Match Bot Simulation:** Data was collected through an automated bot performing 1100 simulated game sessions.

*Note: The Number Match bot simulation data represents a statistically robust sample size (n=1100) compared to the manual testing of the Sumlink App (n=10) and Web App (n=6). Comparisons should be interpreted with this discrepancy in mind.*

## 3. Raw Data Summary

Sumlink App Results (n=10)

| Time (sec) | Moves | Rows Added | Result |
|------------|-------|------------|--------|
| 95.5       | 23    | 3          | Win    |
| 56.86      | 14    | 2          | Win    |
| 129.12     | 26    | 4          | Win    |
| 33.68      | 14    | 1          | Win    |
| 171.93     | 24    | 3          | Win    |
| 127        | 25    | 4          | Win    |
| 127.8      | 27    | 3          | Win    |
| 111.7      | 28    | 3          | Win    |
| 192        | 26    | 4          | Loss   |
| 167.4      | 20    | 3          | Loss   |

Sumlink Web Results (n=6)

| Time (sec) | Moves | Rows Added | Result |
|------------|-------|------------|--------|
| 53.43      | 19    | 1          | Win    |
| 72.48      | 19    | 2          | Win    |
| 60         | 21    | 2          | Win    |
| 80         | 24    | 1          | Win    |
| 120        | 30    | 1          | Win    |
| 159        | 39    | 3          | Win    |

Number Match Bot Simulation Results (n=1100)

- Total Simulations: 1100
- Win Rate: 74.82%
- Avg Completion Time: 45.9 sec
- Avg Add Rows: 2.07
- Avg Score: 217
- Avg Diversity: 44.1%
- Avg Pair Diversity: 3.7
- Most Common Digits: 8, 5, 7

4. Calculated Metrics Table

| Metric                    | Sumlink App | Sumlink Web | Number Match Bot |
|---------------------------|-------------|-------------|------------------|
| Sample Size               | 10          | 6           | 1100             |
| Win Rate                  | 80.00%      | 100.00%     | 74.82%           |
| Avg Completion Time (sec) | 121.30      | 90.82       | 45.90            |
| Avg Moves per Game        | 22.70       | 25.33       | N/A              |
| Avg Rows Added            | 3.00        | 1.67        | 2.07             |

5. Performance Comparison

The platforms exhibit vastly different performance profiles, heavily influenced by the nature of the testing (manual vs. automated). The **Number Match Bot** shows an impressive average completion time of 45.90 seconds, highlighting the efficiency of programmatic execution over human cognition. Between the human-played versions, the **Sumlink Web** version was completed faster on average (90.82 seconds) compared to the **Sumlink App** (121.30 seconds), despite players taking slightly more average moves (25.33 Web vs 22.70 App). This suggests the Web interface may offer quicker interaction mechanisms or that the Web board configurations were marginally easier to parse visually.

6. Gameplay Complexity Comparison

Gameplay complexity can be inferred from the number of moves and the necessity to add rows.

- **Sumlink App:** Demonstrates moderate move complexity (22.70 avg) but requires the most structural assistance (3.00 avg rows added). The higher dependency on row additions indicates a board state that quickly leads to deadlock scenarios.
- **Sumlink Web:** Players needed more average moves (25.33) but required significantly fewer row additions (1.67 avg) to progress. This suggests the Web variant provides a higher density of available starting pairs, allowing for sustained, continuous play without forced board expansion.
- **Number Match Bot:** Exhibits moderate complexity in terms of board expansion (2.07 avg rows added), settling between the Web and App variants. The reported diversity metrics (44.1% diversity, 3.7 pair diversity) suggest the algorithm encounters varied game states.

## 7. Row Addition Analysis

Row additions represent moments where the player (or bot) is unable to find valid moves and must alter the board state.

- The **Sumlink App** heavily relies on this mechanic, averaging 3 additions per game, with some games requiring up to 4. This implies a potentially frustrating user experience where natural combinations frequently stall out.
- The **Sumlink Web** offers a smoother experience, needing only 1.67 additions on average.
- The **Number Match** system balances this mechanic well (2.07 additions), indicating a challenging but solvable default board generation.

## 8. Completion Time Analysis

Completion times reflect both human cognitive load and system efficiency.

- **Sumlink App:** The slowest platform (121.30s avg), with extreme variance ranging from 33.68s to 192s. This volatility suggests high randomness in difficulty.
- **Sumlink Web:** Faster and slightly more consistent (90.82s avg).
- **Number Match Bot:** As expected, the automated bot dramatically outpaces human players (45.90s). The speed underscores the bot's ability to instantly parse the board state.

## 9. Win Rate Analysis

- **Sumlink App (80% Win Rate):** Shows a challenging environment. The losses occurred in games with high time investment (167.4s and 192s) and high row additions (3 and 4), suggesting a failure state where the board becomes completely unmanageable.
- **Sumlink Web (100% Win Rate):** In the small sample size tested, the game was entirely solvable, correlating with the lower requirement for row additions.
- **Number Match (74.82% Win Rate):** The large-scale simulation reveals the true difficulty of the deterministic system. A 75% win rate represents an optimal "casual puzzle" difficulty curve—winnable the vast majority of the time, but punishing enough to retain player engagement.

## 10. Statistical Reliability Discussion

A critical factor in this evaluation is the statistical reliability of the data sources.

- The **Number Match** data (n=1100) provides a highly reliable, statistically significant baseline for how the core game logic performs under stress.
- The **Sumlink App** (n=10) and **Sumlink Web** (n=6) datasets are statistically insignificant and highly prone to human bias, skill variance, and small sample anomalies. While they provide qualitative insights into the user experience, quantitative comparisons between the app and web versions must be treated as preliminary trends rather than definitive conclusions.

## 11. Strengths and Weaknesses of Each System

### Sumlink Android App

- **Strengths:** High difficulty ceiling provides a challenging experience.
- **Weaknesses:** High reliance on row additions indicates frustrating deadlocks. High time variance points to inconsistent generation.

### Sumlink Web App

- **Strengths:** Smooth gameplay flow with fewer dead ends (low row additions). Faster human completion times.
- **Weaknesses:** Potentially too easy (100% win rate in testing). Small sample size masks true statistical profile.

### Number Match Bot (Deterministic)

- **Strengths:** Statistically proven difficulty curve (75% win rate). Excellent data tracking capabilities (diversity metrics, common digits). Highly robust testbed.
- **Weaknesses:** Lacks human usability data; bot efficiency does not directly map to user satisfaction.

## 12. Technical Architecture Comparison

While source code is not fully detailed here, the testing methodologies imply architectural differences. The presence of a headless bot capable of running 1100 simulations indicates that the **Number Match** architecture successfully decouples the game logic from the UI presentation layer. This allows for rapid automated testing and deterministic evaluation. Conversely, the manual testing required for the Sumlink systems suggests a tighter coupling between logic and interface, typical in rapid mobile/web prototyping, which limits automated QA.

## 13. Deterministic vs Non-Deterministic Generation Analysis

The "Deterministic Version" of the Number Match game ensures reproducible board states and predictable difficulty curves. By strictly controlling the generation, the system ensures a balanced challenge (as evidenced by the 74.82% win rate). The high variance in the Sumlink App (times ranging from 33s to 192s, row additions from 1 to 4) strongly suggests a non-deterministic, highly randomized generation approach that occasionally produces trivially easy or unfairly difficult boards.

## 14. Final Ranking

1. **Number Match Puzzle Game:** Best overall architecture and predictable difficulty scaling. The decoupled logic enables robust QA and analytics.

2. **Sumlink Web Application:** Best human user experience. Provides a smoother gameplay loop with fewer frustrations (deadlocks) than the mobile app.
3. **Sumlink Android Application:** Requires the most refinement. The high variance and over-reliance on board expansion tools indicate a need for a better board generation algorithm.

## 15. Conclusion and Future Improvements

The evaluation reveals that while all three implementations provide a functional puzzle experience, the deterministic approach of the Number Match system offers the most balanced and testable framework. The Sumlink mobile app suffers from frustrating board deadlocks, while the Web app provides a smoother but potentially less challenging experience.

### Recommendations for Future Improvement:

- **Implement Deterministic Generation in Sumlink:** Port the deterministic board generation logic from the Number Match system into the Sumlink App and Web platforms to stabilize the difficulty curve.
- **Decouple Logic:** Ensure all future iterations maintain a strict separation between game logic and UI to allow for automated bot testing across all platforms.
- **Scale Human Testing:** Conduct large-scale beta testing with real users on the Number Match deterministic boards to ensure the 75% bot win rate translates to an enjoyable human difficulty level.
- **Adjust Row Addition Economics:** Consider penalizing players for using the "Add Row" feature or ensuring the starting board guarantees more moves before requiring an addition.